

# Kaihua Qin

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## CURRENT POSITIONS

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### Assistant Professor

Department of Computer Science  
University of Warwick

September 2025 – present

### Co-founder

Decentralized Intelligence AG  
<https://d23e.ch>

December 2022 – present

## EDUCATION

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### PhD in Computer Science

Imperial College London

2019 – 2024

London, United Kingdom

Advisor: Prof. Arthur Gervais

Thesis: [Blockchain and Decentralized Finance Security: Incentives, Attacks, and Mitigations](#)

### MSc in Communications and Signal Processing

Imperial College London

2014 – 2015

London, United Kingdom

### BE in Information Engineering

Southeast University

2010 – 2014

Nanjing, China

## RESEARCH

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I work at the intersection of information security, distributed computing, and AI, developing methods to make decentralized systems, such as blockchains, secure, trustworthy, and incentive-compatible. My research spans distributed systems, confidential computing, applied cryptography, and game theory, and applies AI to analyze, protect, and optimize these systems.

I am also interested in the security of AI, focusing on how to keep machine learning models and AI-driven systems robust against adversarial manipulation, privacy breaches, and other emerging threats.

At the heart of my work is a clear goal: to transform rigorous academic research into practical tools and insights that strengthen the digital infrastructure on which society increasingly depends.

## GRANTS AND AWARDS

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- Grants Innosuisse Start-Up Innovation Project (703K CHF) 2024 - 2026  
*AI-Enhanced Smart Contract Auditing*
- Ethereum Foundation Academic Grant (55K USD) 2024  
*Training an EVM-native Generative Pre-trained Transformer*
- The Latest in DeFi Research (30K USD) 2024  
*Fellowship on stablecoin research*

- Ethereum Foundation Academic Grant (50K USD) 2023  
*Dynamic Analysis Framework for EVM*
- Ethereum Foundation Academic Grant (72K USD) 2023  
*Denial-of-Service Implications of Blockchain Censorship*
- Ethereum Foundation Academic Grant (25K USD) 2023  
*Blockchain Censorship — Quantitative Analysis of Censorship on Public Blockchains*
- Cybersecurity Best Practical Paper Award 2023
- Best DeFi Papers Award 2023

## PUBLICATIONS AND PREPRINTS

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Bibliometrics from [Google Scholar](#): citations = 3,538, h-index = 22, i10-index = 26 (as of July 2026).

31. Yang, Sen, **Kaihua Qin**, Aviv Yaish, and Fan Zhang. “Insecurity Through Obscurity: Veiled Vulnerabilities in Closed-Source Contracts”. In *Proceedings of the 2026 ACM SIGSAC Conference on Computer and Communications Security*. ACM, 2026.
30. Wang, Ziyue, Cheuk Wang Maurice Ng, Chenchen Yu, Strick Sheng, **Kaihua Qin**, and Liyi Zhou. “Transferable Self-Evolving Playbooks for Agentic Security Auditing”. *arXiv preprint arXiv:2606.16420* (2026).
29. **Qin, Kaihua**, Dawn Song, and Arthur Gervais. “SCDBench: A Benchmark for LLM-Based Smart Contract Decompilers”. *arXiv preprint arXiv:2605.29059* (2026).
28. Wang, Ziyue, Jiangshan Yu, **Kaihua Qin**, Dawn Song, Arthur Gervais, and Liyi Zhou. “TxRay: Agentic Postmortem of Live Blockchain Attacks”. *arXiv preprint arXiv:2602.01317* (2026).
27. **Qin, Kaihua**, Zhe Ye, Zhun Wang, Weilin Li, Liyi Zhou, Chao Zhang, Dawn Song, and Arthur Gervais. “Enhancing Smart Contract Security Analysis with Execution Property Graphs”. *Proceedings of the ACM on Software Engineering* 2, no. ISSTA (2025): 1101–1122.
26. Guan, Maggie Yongqi, Yaman Yu, Tanusree Sharma, Molly Zhuangtong Huang, **Kaihua Qin**, Yang Wang, and Kanye Ye Wang. “Security Perceptions of Users in Stablecoins: Advantages and Risks within the Cryptocurrency Ecosystem”. In *2025 IEEE Symposium on Security and Privacy (SP)*, 2753–2771. IEEE, 2025.
25. Tsuchiya, Taro, Liyi Zhou, **Kaihua Qin**, Arthur Gervais, and Nicolas Christin. “Blockchain Amplification Attack”. *Proceedings of the ACM on Measurement and Analysis of Computing Systems* 9, no. 1 (2025): 1–30.
24. Potter, Yujin, Kornrapat Pongmala, **Kaihua Qin**, Arian Klages-Mundt, Philipp Jovanovic, Christine Parlour, Arthur Gervais, and Dawn Song. “What Drives the (In) stability of a Stablecoin?” In *2024 IEEE International Conference on Blockchain and Cryptocurrency (ICBC)*, 316–324. IEEE, 2024.
23. Wahrstätter, Anton, Jens Ernstberger, Aviv Yaish, Liyi Zhou, **Kaihua Qin**, Taro Tsuchiya, Sebastian Steinhorst, Davor Svetinovic, Nicolas Christin, and Mikolaj Barczentewicz. “Blockchain censorship”. In *Proceedings of the ACM Web Conference 2024*, 1632–1643. 2024.

22. Gan, Rundong, Liyi Zhou, Le Wang, **Kaihua Qin**, and Xiaodong Lin. “Defialigner: Leveraging symbolic analysis and large language models for inconsistency detection in decentralized finance”. In *6th Conference on Advances in Financial Technologies (AFT 2024)*, 7–1. Schloss Dagstuhl–Leibniz-Zentrum für Informatik, 2024.
21. Yaish, Aviv, **Kaihua Qin**, Liyi Zhou, Aviv Zohar, and Arthur Gervais. “Speculative {Denial-of-Service} Attacks In Ethereum”. In *33rd USENIX security symposium (USENIX Security 24)*, 3531–3548. 2024.
20. David, Isaac, Liyi Zhou, **Kaihua Qin**, Dawn Song, Lorenzo Cavallaro, and Arthur Gervais. “Do you still need a manual smart contract audit?” *arXiv preprint arXiv:2306.12338* (2023).
19. Wahrstätter, Anton, Liyi Zhou, **Kaihua Qin**, Davor Svetinovic, and Arthur Gervais. “Time to bribe: Measuring block construction market”. *arXiv preprint arXiv:2305.16468* (2023).
18. Zhou, Liyi, Xihan Xiong, Jens Ernstberger, Stefanos Chaliasos, Zhipeng Wang, Ye Wang, **Kaihua Qin**, Roger Wattenhofer, Dawn Song, and Arthur Gervais. “Sok: Decentralized finance (defi) attacks”. In *2023 IEEE Symposium on Security and Privacy (SP)*, 2444–2461. IEEE, 2023.
17. Feng, Ding, Rupert Hitsch, **Kaihua Qin**, Arthur Gervais, Roger Wattenhofer, Yaxing Yao, and Ye Wang. “Defi auditing: Mechanisms, effectiveness, and user perceptions”. In *International Conference on Financial Cryptography and Data Security*, 320–336. Springer, 2023.
16. Wang, Zhipeng, Stefanos Chaliasos, **Kaihua Qin**, Liyi Zhou, Lifeng Gao, Pascal Berrang, Benjamin Livshits, and Arthur Gervais. “On how zero-knowledge proof blockchain mixers improve, and worsen user privacy”. In *Proceedings of the ACM Web Conference 2023*, 2022–2032. 2023.
15. Gai, Yu, Liyi Zhou, **Kaihua Qin**, Dawn Song, and Arthur Gervais. “Blockchain large language models”. *arXiv preprint arXiv:2304.12749* (2023).
14. Raun, Christoffer, Benjamin Estermann, Liyi Zhou, **Kaihua Qin**, Roger Wattenhofer, Arthur Gervais, and Ye Wang. “Leveraging machine learning for bidding strategies in miner extractable value (mev) auctions”. *Cryptology ePrint Archive* (2023).
13. Yaish, Aviv, Maya Dotan, **Kaihua Qin**, Aviv Zohar, and Arthur Gervais. “Suboptimality in defi”. *Cryptology ePrint Archive* (2023).
12. **Qin, Kaihua**, Stefanos Chaliasos, Liyi Zhou, Benjamin Livshits, Dawn Song, and Arthur Gervais. “The blockchain imitation game”. In *32nd USENIX Security Symposium (USENIX Security 23)*, 3961–3978. 2023.
11. **Qin, Kaihua**, Jens Ernstberger, Liyi Zhou, Philipp Jovanovic, and Arthur Gervais. “Mitigating decentralized finance liquidations with reversible call options”. In *International Conference on Financial Cryptography and Data Security*, 344–362. Springer, 2023.
10. Wang, Zhipeng, **Kaihua Qin**, Duc Vu Minh, and Arthur Gervais. “Speculative multipliers on defi: Quantifying on-chain leverage risks”. In *International Conference on Financial Cryptography and Data Security*, 38–56. Springer, 2022.
9. **Qin, Kaihua**, Liyi Zhou, and Arthur Gervais. “Quantifying blockchain extractable value: How dark is the forest?” In *2022 IEEE Symposium on Security and Privacy (SP)*, 198–214. IEEE, 2022.

8. **Qin, Kaihua**, Liyi Zhou, Pablo Gamito, Philipp Jovanovic, and Arthur Gervais. "An empirical study of defi liquidations: Incentives, risks, and instabilities". In *Proceedings of the 21st ACM internet measurement conference*, 336–350. 2021.
7. Zhou, Liyi, **Kaihua Qin**, Antoine Cully, Benjamin Livshits, and Arthur Gervais. "On the just-in-time discovery of profit-generating transactions in defi protocols". In *2021 IEEE Symposium on Security and Privacy (SP)*, 919–936. IEEE, 2021.
6. Zhou, Liyi, **Kaihua Qin**, Christof Ferreira Torres, Duc V Le, and Arthur Gervais. "High-frequency trading on decentralized on-chain exchanges". In *2021 IEEE Symposium on Security and Privacy (SP)*, 428–445. IEEE, 2021.
5. **Qin, Kaihua**, Liyi Zhou, Yaroslav Afonin, Ludovico Lazzaretti, and Arthur Gervais. "CeFi vs. DeFi—Comparing Centralized to Decentralized Finance". *arXiv preprint arXiv:2106.08157* (2021).
4. Zhou, Liyi, **Kaihua Qin**, and Arthur Gervais. "A2mm: Mitigating frontrunning, transaction re-ordering and consensus instability in decentralized exchanges". *arXiv preprint arXiv:2106.07371* (2021).
3. **Qin, Kaihua**, Liyi Zhou, Benjamin Livshits, and Arthur Gervais. "Attacking the defi ecosystem with flash loans for fun and profit". In *International conference on financial cryptography and data security*, 3–32. Springer, 2021.
2. Janin, Simon, **Kaihua Qin**, Akaki Mamageishvili, and Arthur Gervais. "Filebounty: Fair data exchange". In *2020 IEEE European Symposium on Security and Privacy Workshops (EuroS&PW)*, 357–366. IEEE, 2020.
1. **Qin, Kaihua**, Henryk Hadass, Arthur Gervais, and Joel Reardon. "Applying private information retrieval to lightweight bitcoin clients". In *2019 Crypto Valley Conference on Blockchain Technology (CVCBT)*, 60–72. IEEE, 2019.

## PROFESSIONAL SERVICES

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### Program Committee Chairs

- ACM CCS Workshop on Decentralized Finance and Security 2024 (co-chair with Liyi Zhou)
- ACM CCS Workshop on Decentralized Finance and Security 2023 (co-chair with Fan Zhang)

### Program Committee Members

- ACM Conference on Computer and Communications Security (CCS '24, '25, '26)
- IEEE Symposium on Security and Privacy (S&P '25, '27)
- USENIX Security Symposium (Security '25, '27)
- The Network and Distributed System Security Symposium (NDSS '25, '26, '27)
- IEEE European Symposium on Security and Privacy (EuroS&P '26)
- International Conference on Financial Cryptography and Data Security (FC '24, '25, '26, '27)
- ACM ASIA Conference on Computer and Communications Security (ASIACCS '27)
- The international conference on Advances in Financial Technologies (AFT '24, '25, '26)

- Nanyang Blockchain Conference (NBC '26)
- IEEE International Conference on Blockchain and Cryptocurrency (ICBC '25)
- ACM Workshop on Decentralized Finance and Security (DeFi '21, '22, '25, '26)
- International Workshop on Formal Methods for Blockchains (FMBC '26)
- Designing DeFi Conference (D<sup>2</sup> '26)
- Web3 Scholars Conference (WSC '26)
- Crypto Economics Security Conference (CESC '22)

## Reviewer

- IEEE Transactions on Information Forensics and Security (TIFS)
- IEEE Transactions on Dependable and Secure Computing (TDSC)
- ACM Transactions on Privacy and Security (TOPS)
- ACM Transactions on Software Engineering and Methodology (TOSEM)
- ACM Transactions on the Web (TWEB)
- Distributed Ledger Technologies: Research and Practice (DLT)
- Blockchain: Research and Applications (BCRA)
- IEEE Access
- Future Generation Computer Systems
- Financial Innovation
- Electronic Commerce Research
- ACM CHI conference on Human Factors in Computing Systems (CHI '24, '26)
- IEEE International Conference on Distributed Computing Systems (ICDCS '23)
- The Conference on Web and Internet Economics (WINE '22)
- International Conference on Information Systems (ICIS '22)
- The Web Conference (WWW '21)

## EXPERIENCE

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### Affiliated Researcher

Yale University

Host: Prof. Fan Zhang and Prof. Zhong Shao

February 2024 – August 2025

### Visiting Researcher

University of California, Berkeley

Host: Prof. Dawn Song

May 2022 – August 2022

### Research Intern

Chainlink Labs

February 2022 – April 2022

### Software Engineer

Cisco

March 2016 – August 2018

*Updated July 2026*